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CloudNativeCon

China 2025





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An Alternative Metadata System for Large Kubernetes Clusters

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Contents



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- 01/ Background
- 02/ System Design
- 03/ Optimization
- 04/ Performace
- 05/ OpenSource



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01 / Background

01/ Kubernetes in ByteDance

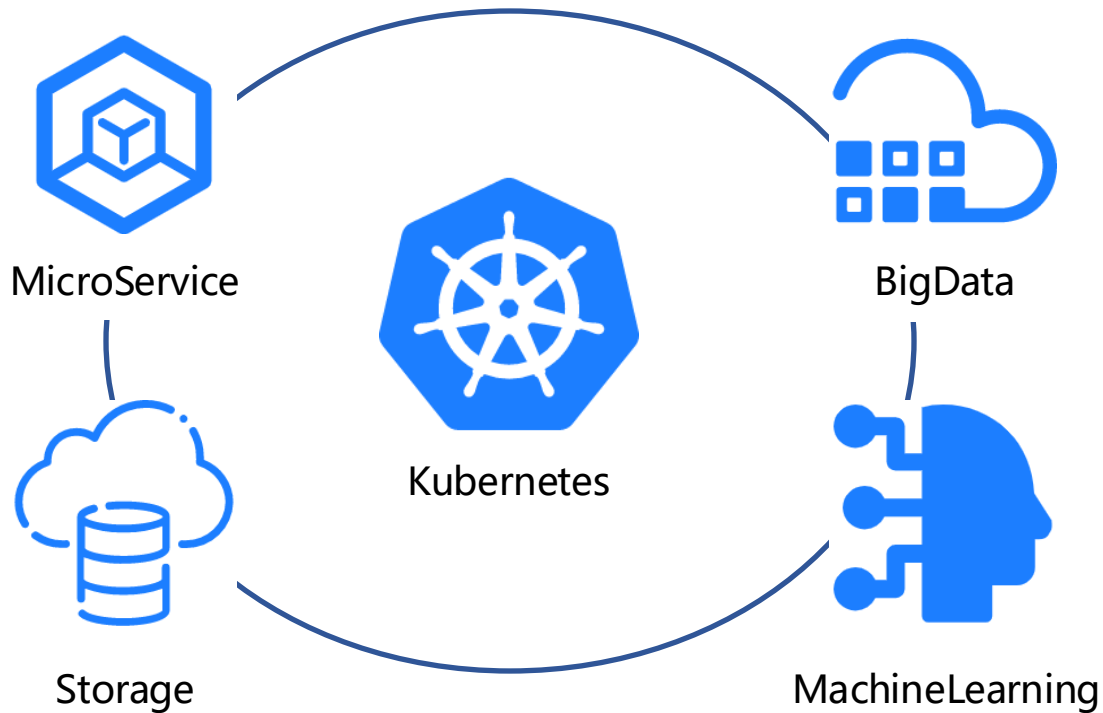


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01/ Scale Up Kubernetes Cluster



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Horizontal



Vertical



01/ Centralized architecture in Kubernetes



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All K8s components
communicate through kube-apiserver



All metadata in K8s
are stored in etcd

01/ Centralized architecture in Kubernetes

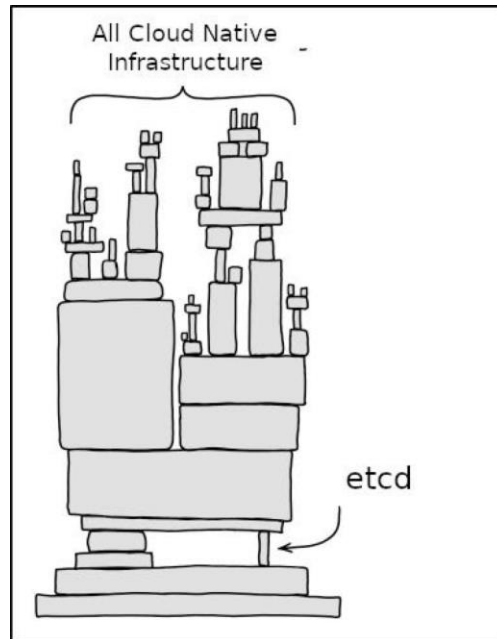
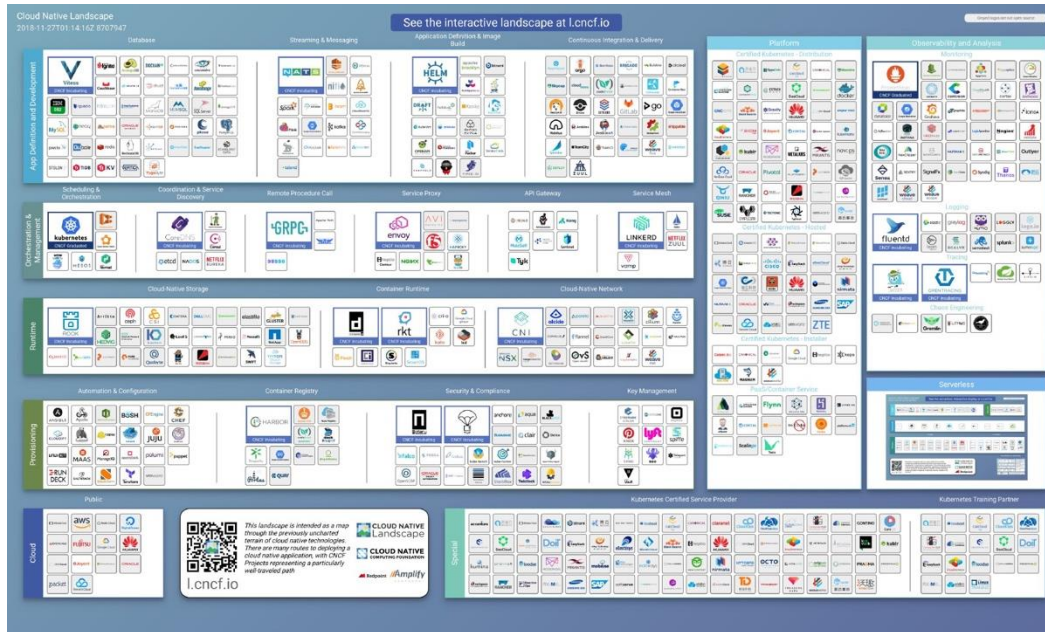


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etcd becomes the **main bottlenecks** of the system in large k8s cluster

01/ Architecture and Issues of etcd3

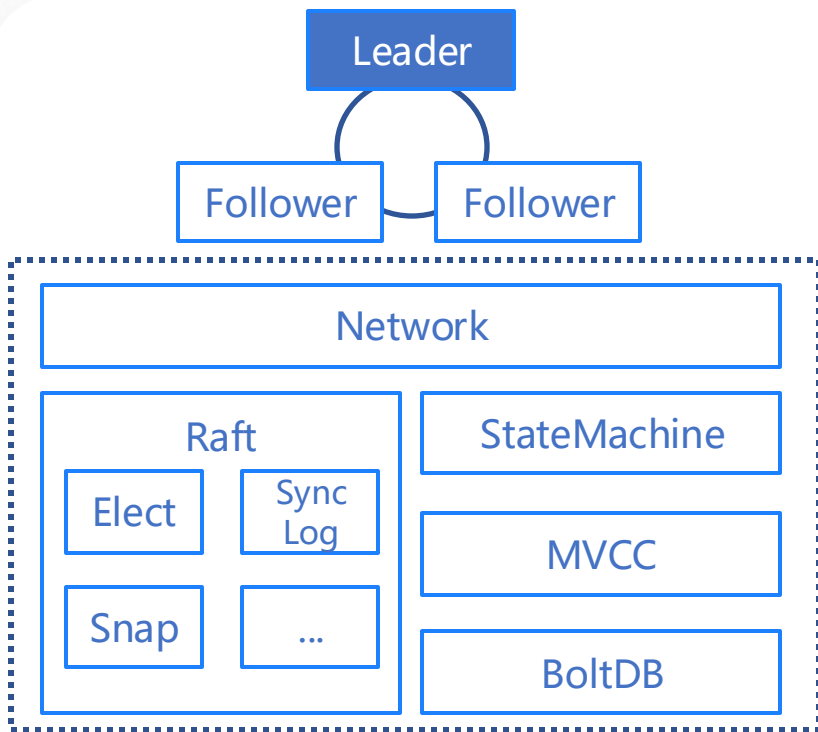


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issues in large clusters

- › Bad expensive read performance
- › Expensive read cause OOM
- › Single raft group limits txn throughput
- › Serial apply limits txn throughput
- › Jitter when snapshot or defrag
- › Jitter when events are expired
- › ...

01/ Optimize K8s Metadata System



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Tune etcd
Split data by resource
...



New metadata system
only for kubernetes



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02/System Design

02/ Requirement K8s Metadata System

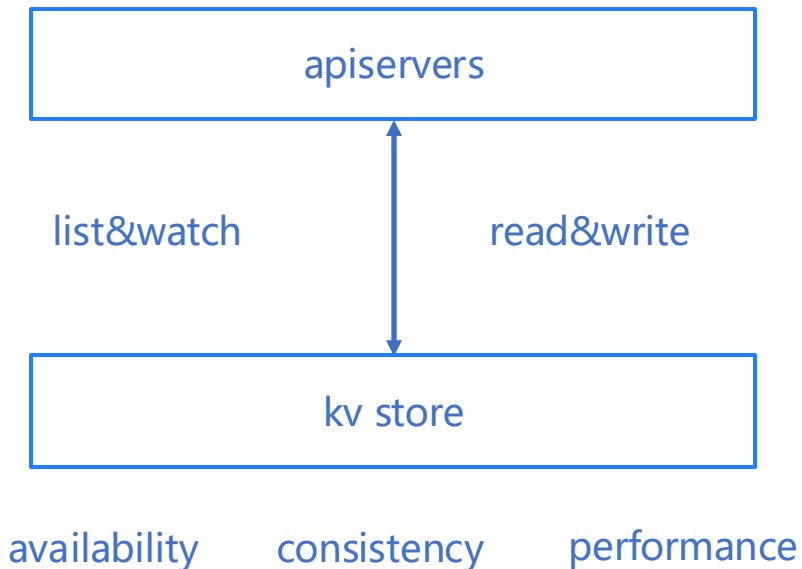


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Version Control

- › Expose KV Version (resourceVersion)

Write

- › Support Create/Update/Delete
- › CAS with resourceVersion
- › TTL for k8s events(Rough&Renew&Const)

Read

- › Snapshot List at resourceVersion
- › Get KV with resourceVersion

Watch

- › Watch order changes from resourceVersion
- › ...

02/ KubeBrain: New Metadata System Inspired by Kine



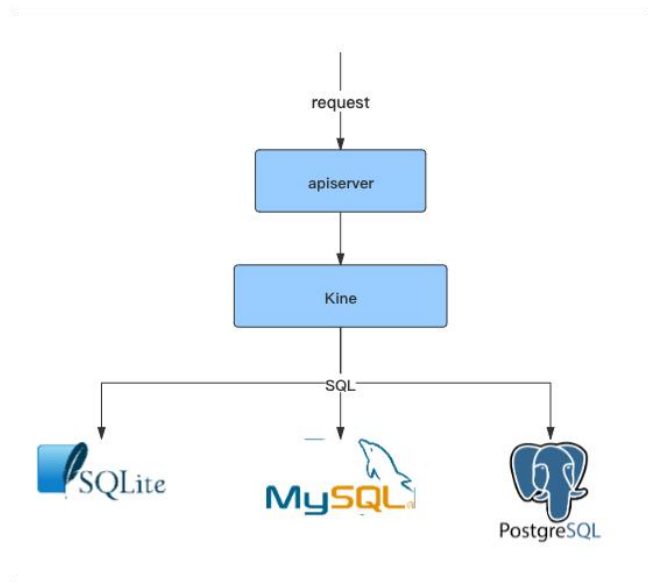
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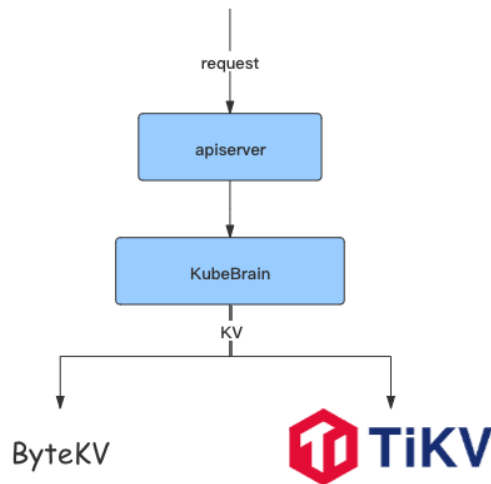
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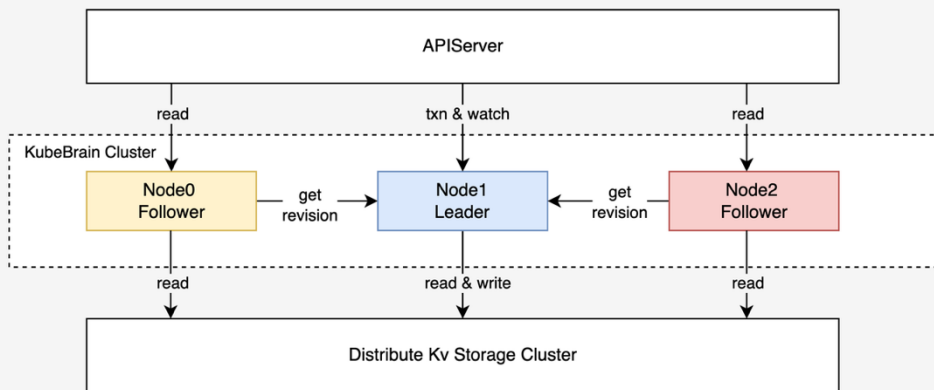
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Kine



KubeBrain





Share-Storage

- › Stateless
- › Share distribute kv storage to persist data

Leader-Follower

- › Leader process Txn & Watch
- › Follower process Get & List

02/ Layers



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Client Layer

Access Layer

Logic Layer

Storage Layer



02/ Storage Layer - Interface and Adaptors

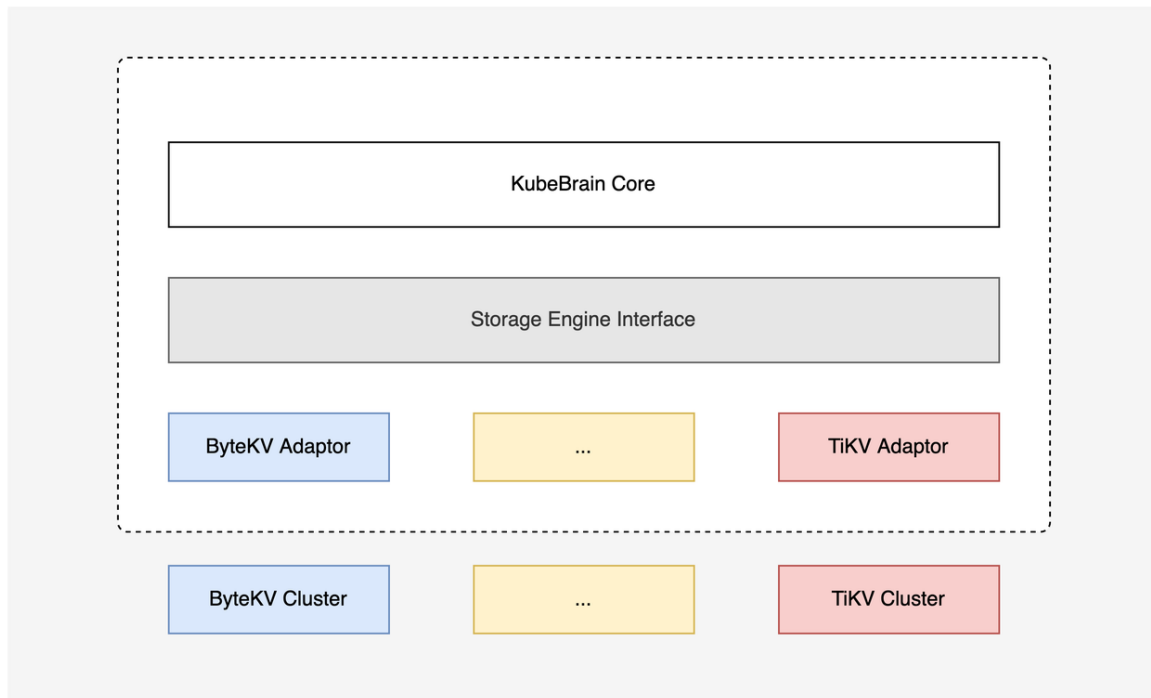


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02/ Storage Layer - ByteKV

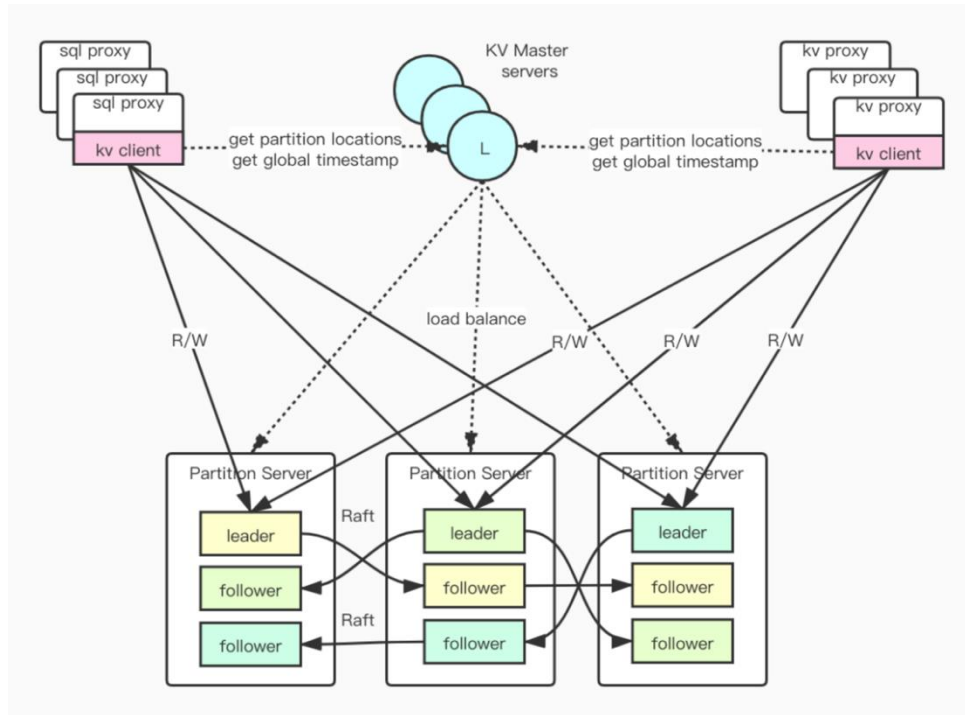


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- › Sort by alphabetical order
- › Multi-raft group
- › Expose global HLC
- › Support write txn
- › Support iterator
- › Linearizable
- › Snapshot Isolation
- › High performance
- › Custom partition splitting rule

02/ Layers

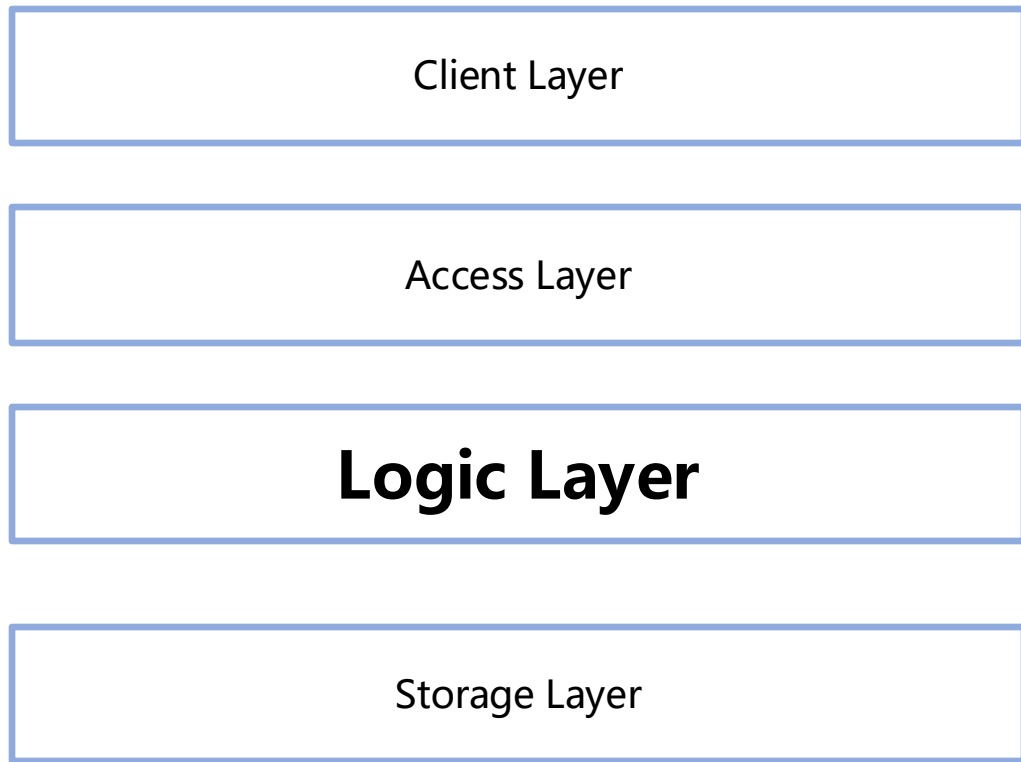


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Client Layer

Access Layer

Logic Layer

Storage Layer

02/ Logic Layer - Election

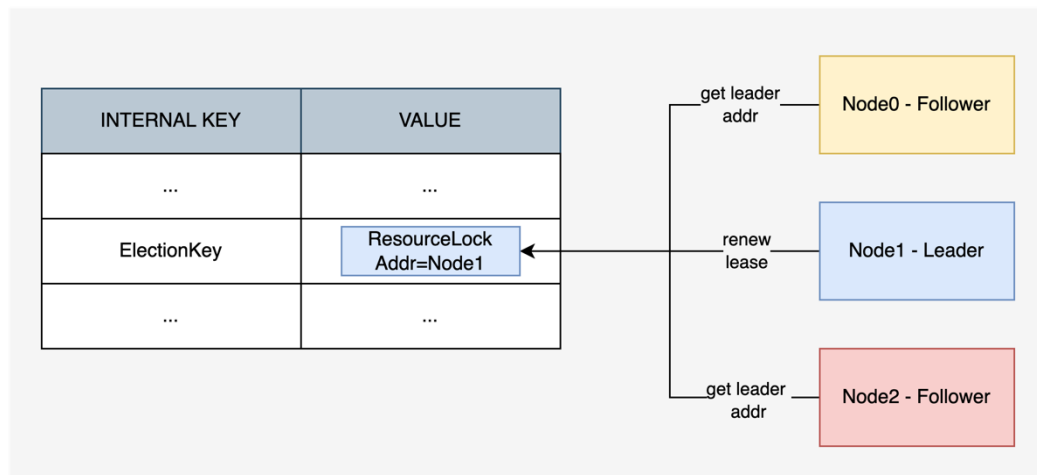


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- › Election based on Lease & CAS (K8s Resource Lock)
- › Store lease in shared KV store
- › Find leader from lease content
- › Maintain Logical TSO in mem of leader to deal revision

02/ Logic Layer - Data Model (MVCC)



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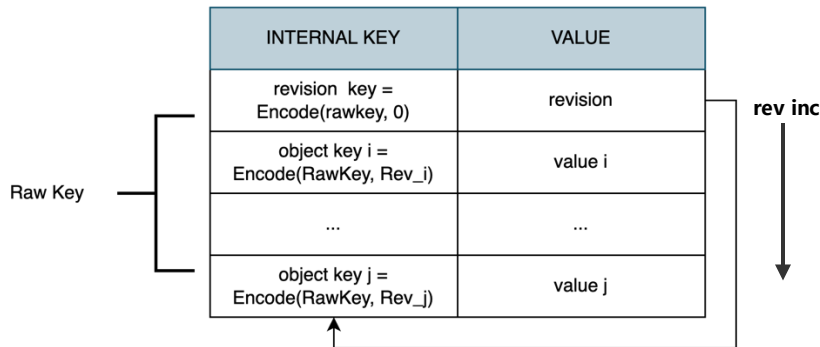


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Big-endian
Encode



Encode Key

- > **Object Key:** store the multiple versions of data
Encode(RawKey, Rev_i)
- > **Revision Key:** store the latest revision
Encode(RawKey, 0)

Characteristic

- > Order of internal keys depend on **raw key** & **rev**
- > Internal keys for the same raw key are **adjacent**
- > Easy to seek/get/list
- > No hot key

02/ Logic Layer – Write Transaction



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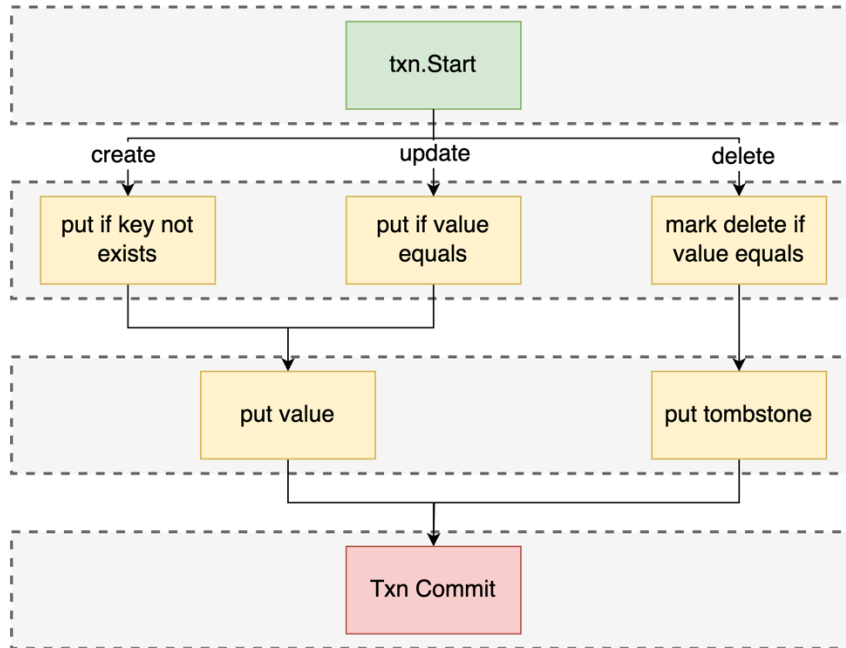


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**CAS
Revision
Key**

**Add
Object Key**



02/ Logic Layer – Watch



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Mathias Verraes

@mathiasverraes

Follow



There are only two hard problems in distributed systems: 2. Exactly-once delivery
1. Guaranteed order of messages 2. Exactly-once delivery

11:40 AM - 14 Aug 2015

7,233 Retweets 5,793 Likes



71



7.2K



5.8K

02/ Logic Layer – Watch

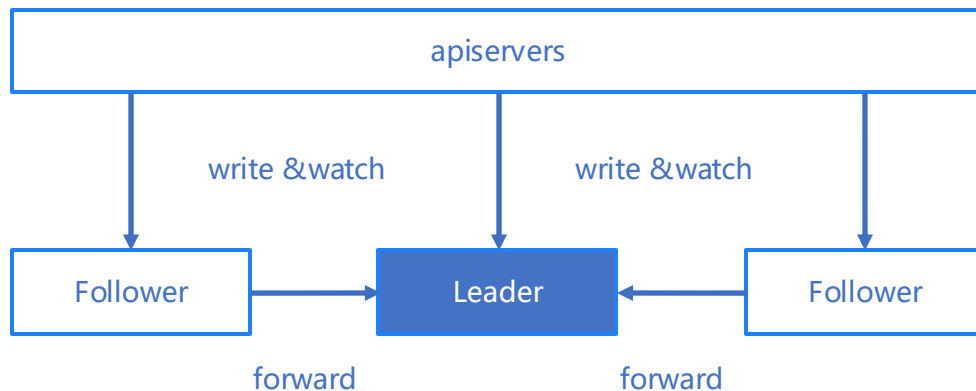


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1 Leader + N Followers

- > **Leader:** process write and watch
- > **Follower:** read only and forward write & watch to leader

02/ Logic Layer – Watch

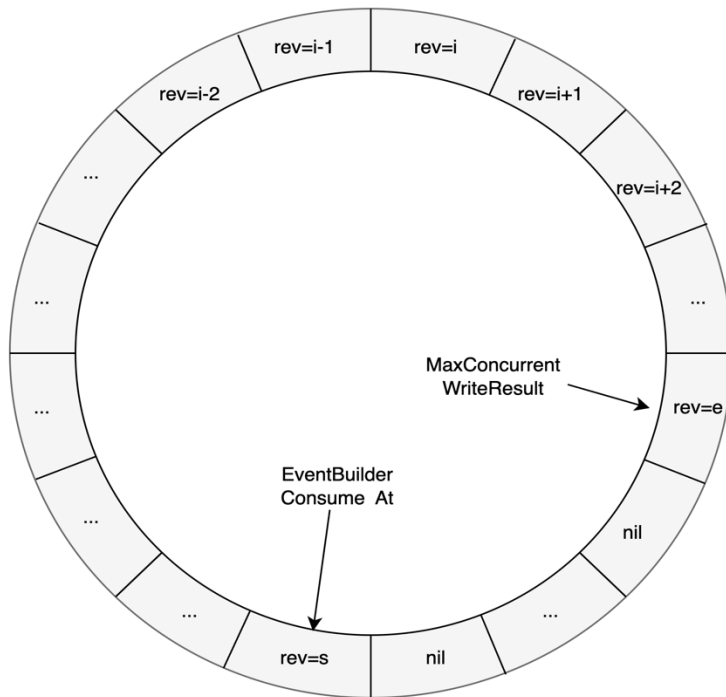


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- > **Use Ring Buffer to record concurrent write results**
- > **Use Rev as index**
- > **Consume ordered results to build ordered events**

02/ Logic Layer – Watch

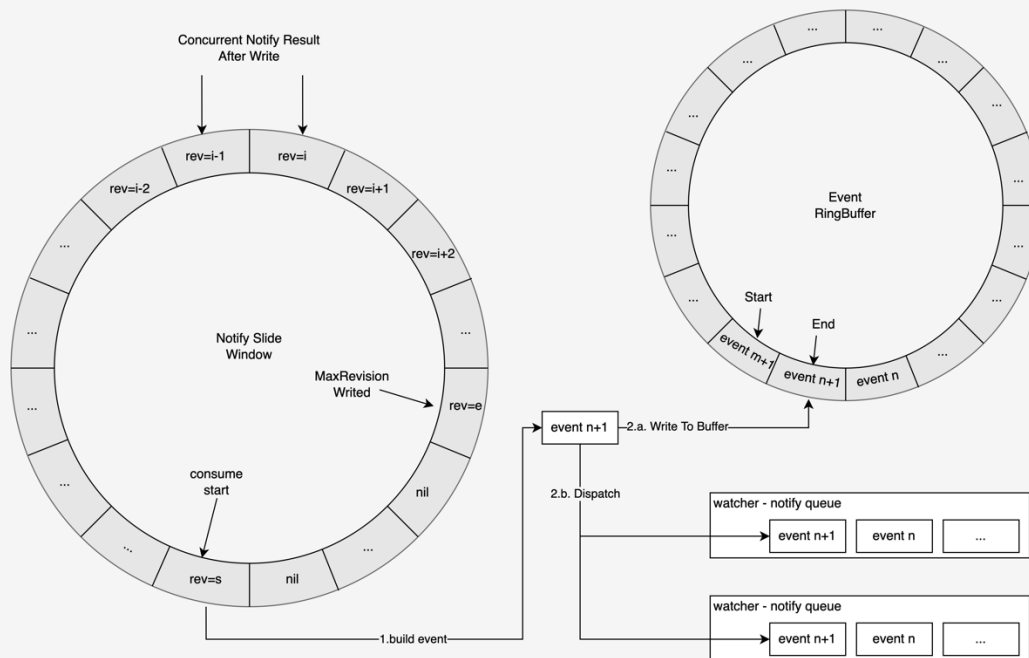


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02/ Logic Layer – Watch

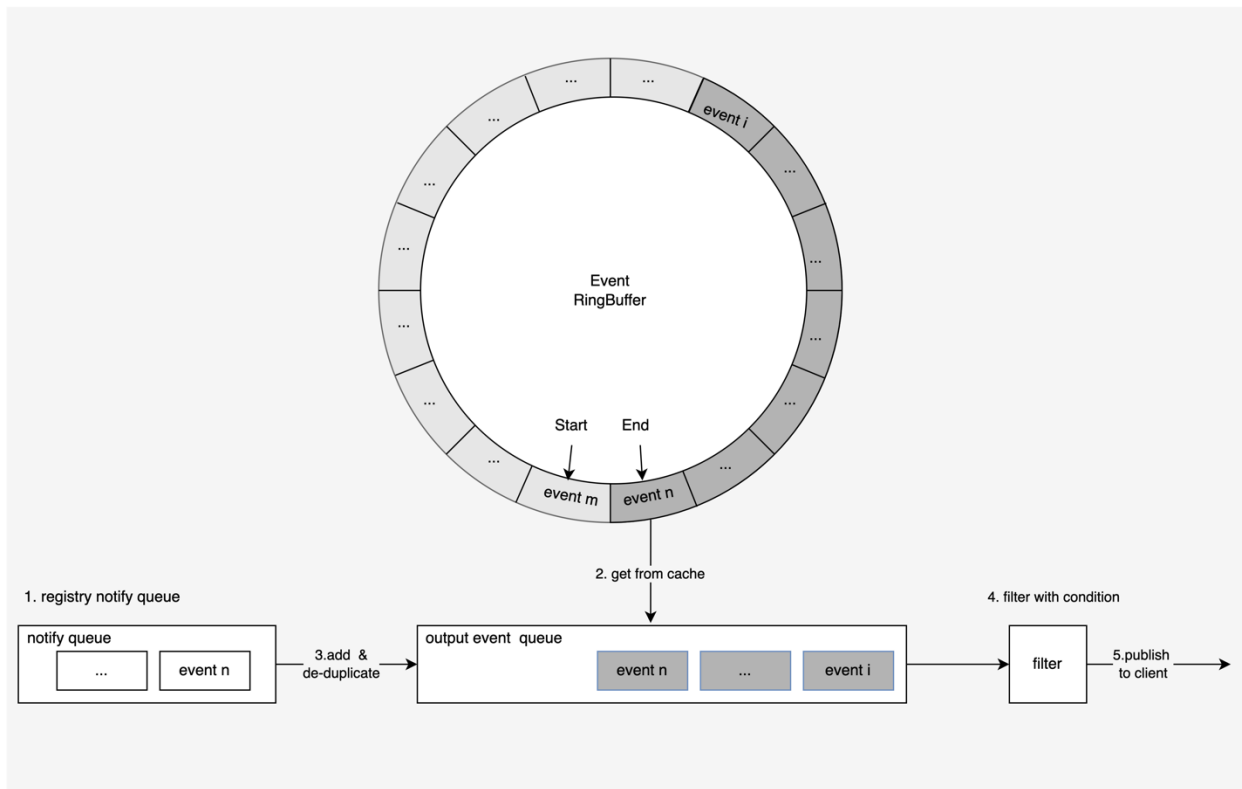


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02/ Logic Layer – Get

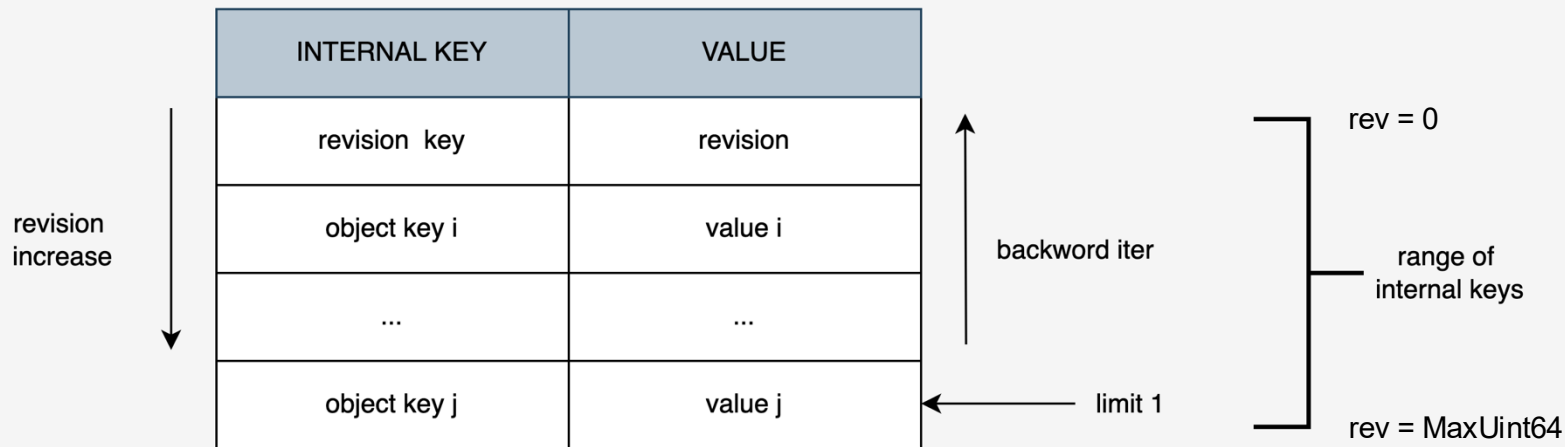


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02/ Logic Layer – List

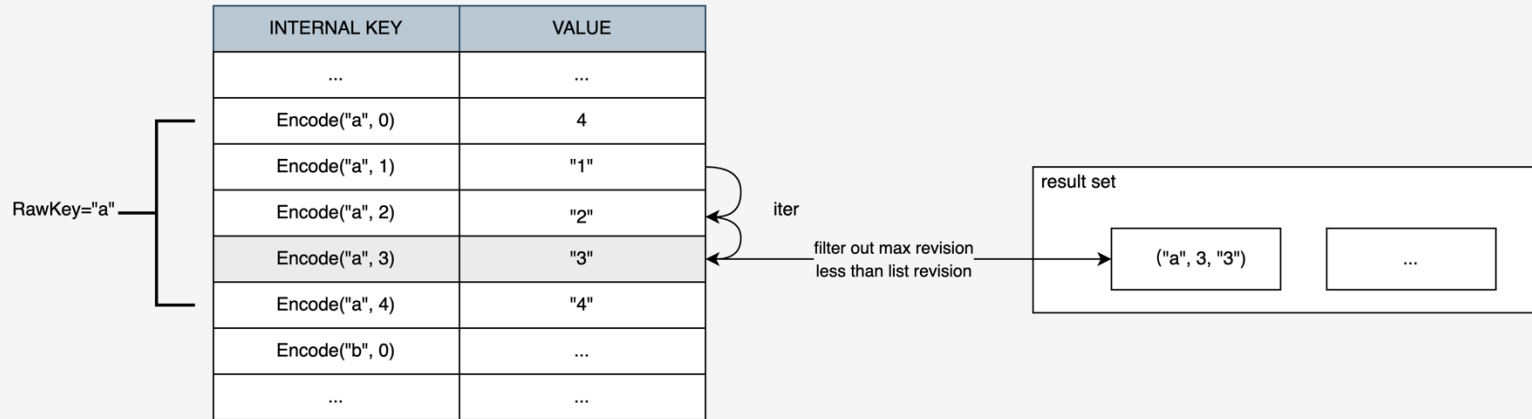


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02/ Layers



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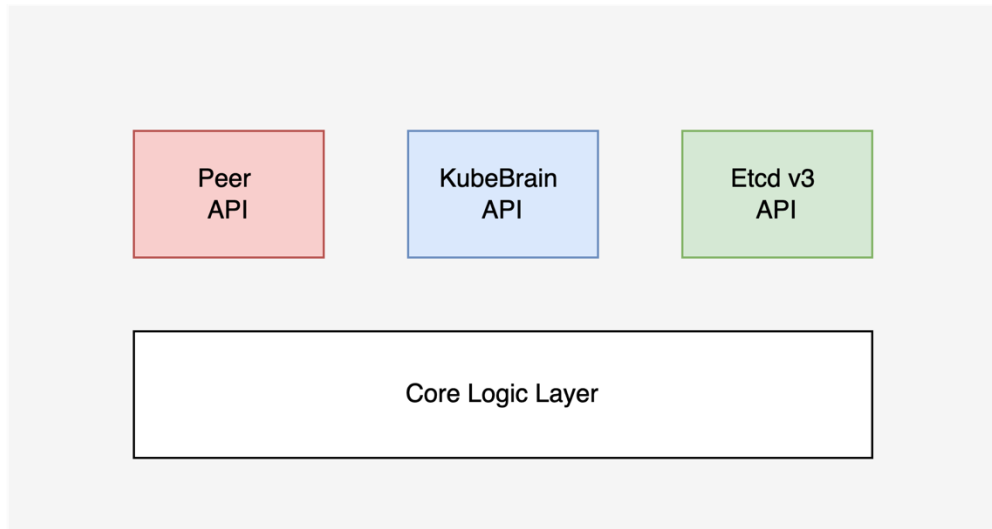
Client Layer

Access Layer

Logic Layer

Storage Layer





Peer API: for peer communication

- › For peer communication

Etcd API :

- › For etcdv3 client
- › Impl subset of etcd v3 API
- › Compatible for k8s < 1.25

KubeBrain API

- › For kubebrain client
- › Support efficient APIs



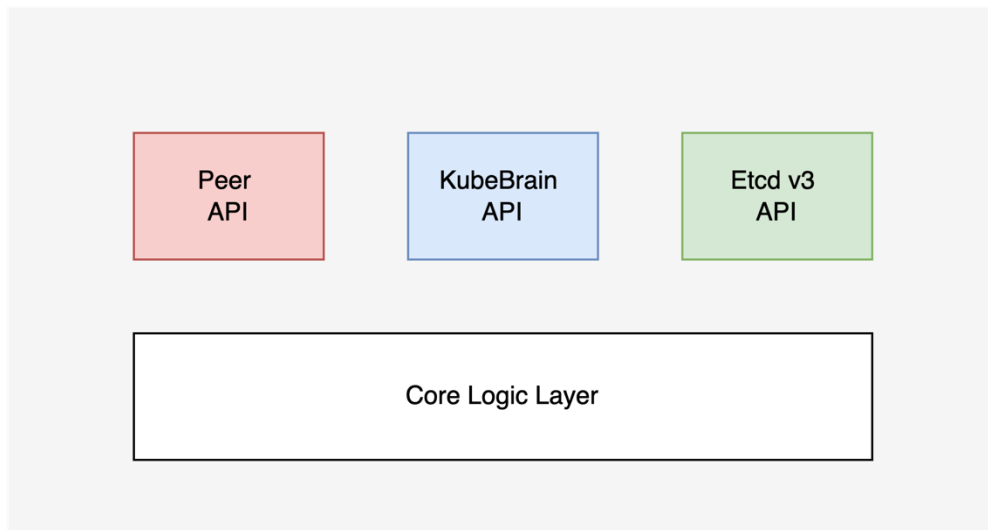
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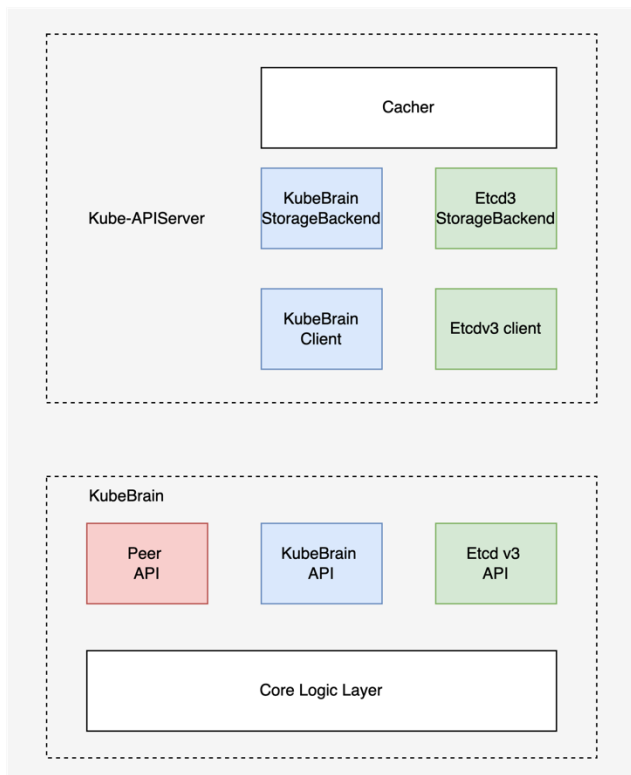
Client Layer

Access Layer

Logic Layer

Storage Layer





More Efficient Storage Backend & SDK

- › RW Separate
- › Stream List
- › Load Balance



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03/Optimization

03/ Optimize Write Txn

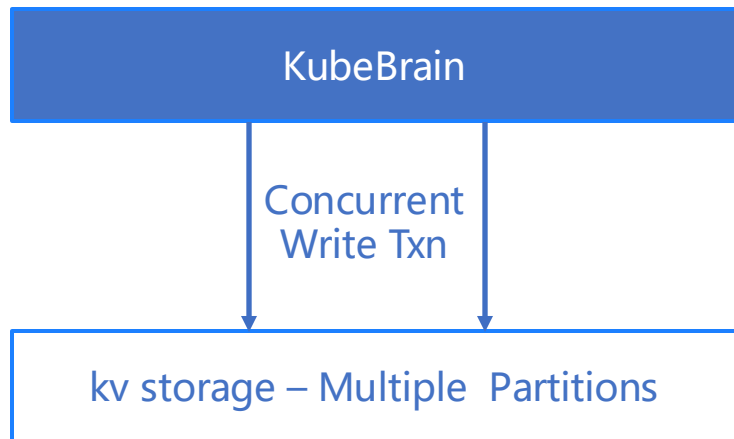


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Concurrent Write Txn

- › Optimistic concurrency control
- › No serial applying
- › No single raft group

03/ Optimize Write Txn

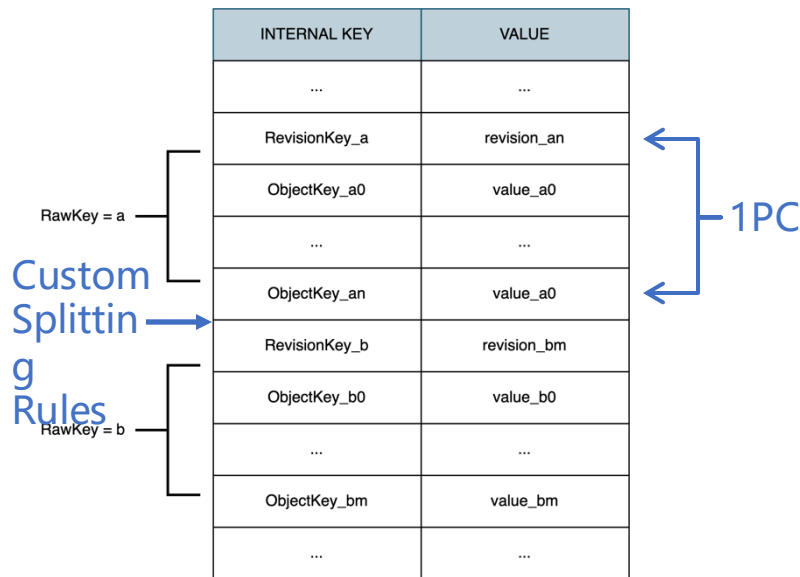


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Op Txn

- > Custom splitting rules to ensure internal keys of a raw key **in the same partition**
- > Txn in one partition impl by 1PC
- > CAS in storage engine to reduce RTT

03/ Optimize Data Fetching



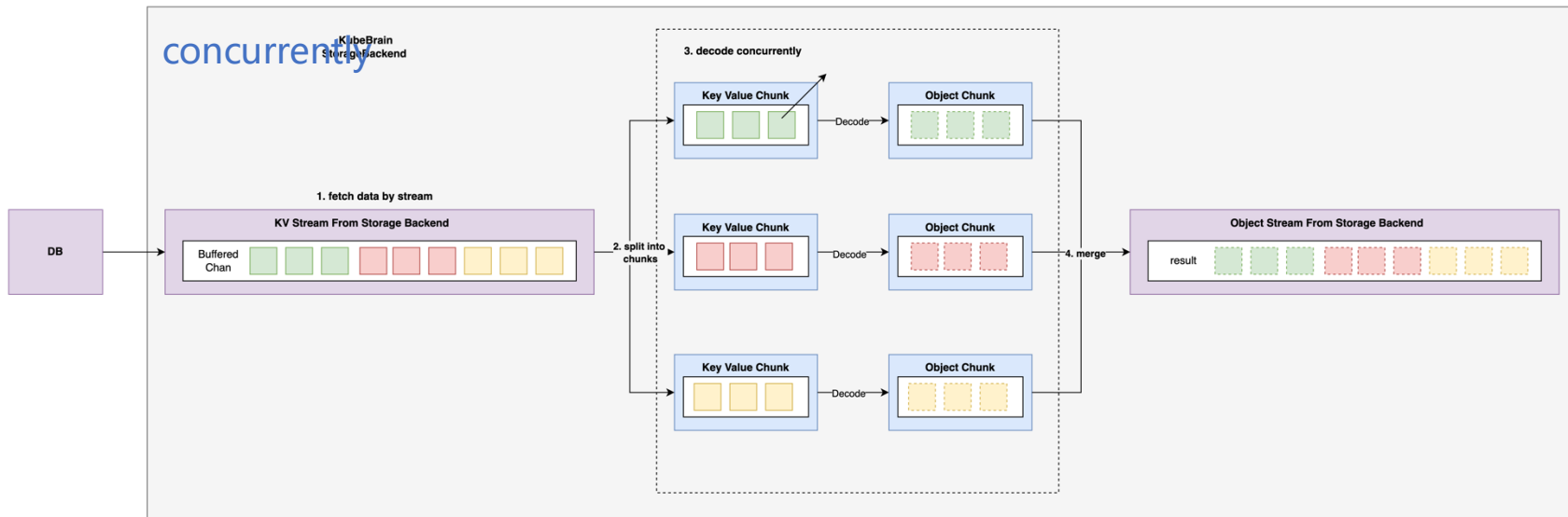
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Fetch data by stream (list & watch) and decode



03/ Optimize Data Fetching



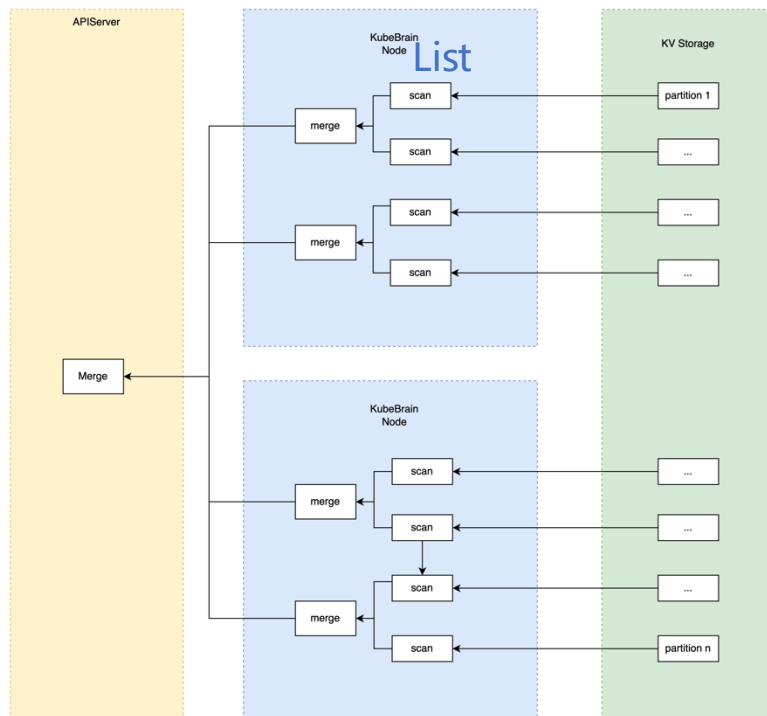
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Partition-level concurrent





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04/Performance

04/ Performance on ByteKV



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Benchmark Env

- › 50k nodes & 2M pods

Production Env

- › 20k nodes & 1M pods
- › Max Total Write QPS > 15k
- › Max Total QPS > 20k
- › PCT99 Write Latency < 1s
- › PCT99 Event E2E Latency < 1s



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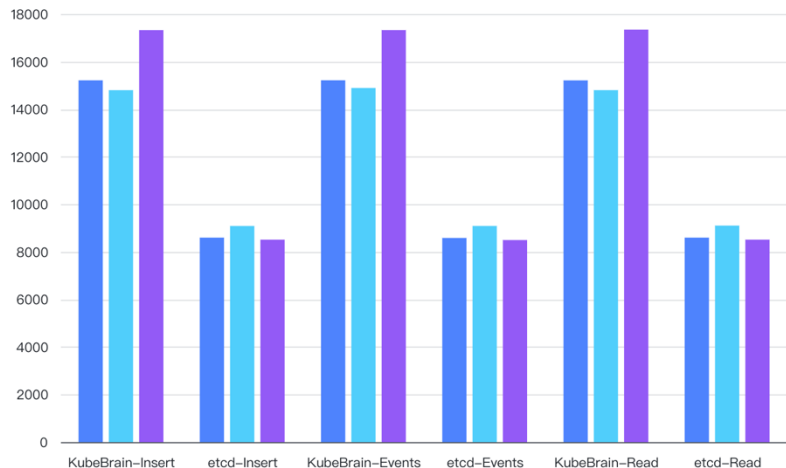
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05/OpenSource



Insert And Read Throughput



Now

- › open source based on TiKV
- › github.com/kubewharf/kubebrain
- › Compatible with k8s < 1.25

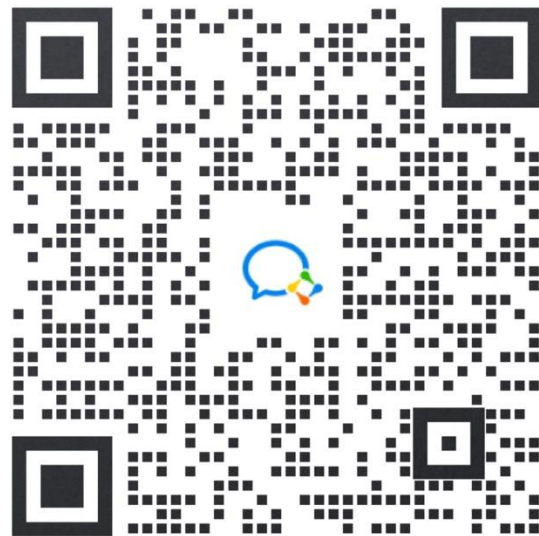
Future

- › More features: bookmark, graceful leader change etc.
- › Performance optimization on TiKV
- › Compatible with latest k8s

Thanks for your listening ~



Describe ByteDance CloudNative



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